Random Order Enables Doubly-Uniform Online Regression

Pahan Dewasurendra
Johns Hopkins University

Abstract

Online square-loss regression admits no sublinear regret bound that is simultaneously independent of the comparator norm and covariate scale in dimension above one when covariates are adaptive. We show that uniformly shuffling only the covariates changes the answer sharply, even if bounded labels remain fully adaptive. The unregularized Vovk–Azoury–Warmuth forecaster has expected regret at most $B^2 \sum_{t \le T} \min(r, t)/t$, where r is the final design rank. The result needs no norm, condition-number, distributional, or ambient-dimension bound. Its core is a distributional theorem for online leverage. Conditional on any fixed covariate multiset, the leverage sum is dominated in Laplace transform by $r + \sum_{t=r+1}^{T} \mathrm{Bernoulli}(r/t)$. This yields Bennett tails, an exact expected rank-profile formula, and kernelization by finite-span reduction. The domination is asymptotically exact in distribution already in one dimension. A Bayesian lower bound proves the minimax rate $\Theta(B^2 r \log(eT/r))$ and recovers leading constant one in the large-block regime. Finally, entropy transport quantifies degradation under imperfect shuffling. The result isolates exchangeable covariate arrival as enough to recover the transductive logarithmic rate without randomizing the labels or revealing the final Gram matrix.

1 INTRODUCTION

In online linear regression, the learner observes a covariate x_t, predicts $\widehat{y}_t$, and then sees a bounded label $y_t \in [-B, B]$. Its square-loss regret to $u \in \mathbb{R}^d$ is

$$\mathrm{Reg}_T(u) = \sum_{t=1}^{T} (\widehat{y}_t - y_t)^2 - \sum_{t=1}^{T} (\langle u, x_t \rangle - y_t)^2.$$

A bound is *doubly uniform* if it depends on neither $\|u\|$ nor the scale of the covariates. This is a demanding but natural invariance: rescaling a feature and inversely rescaling its coefficient should not change the statistical problem.

The sequential and transductive answers differ. If the final Gram matrix is known in advance, logarithmic doubly-uniform regret is possible with the optimal leading term $B^2 d \log T$ (Bartlett et al., 2015; Gaillard et al., 2019). With sequentially revealed covariates, Chen et al. (2026) recently proved a dimension-dependent phase transition. The one-dimensional problem has logarithmic regret, while every dimension above one permits linear regret under adaptive covariates. Their smooth-density condition recovers a $\widetilde{O}(B^2 \sqrt{d C_{\mathrm{cov}} T})$ guarantee, including the i.i.d. endpoint $C_{\mathrm{cov}} = 1$.

We identify a substantially sharper endpoint. Fix any labeled multiset of covariates and reveal a uniform random permutation. Only the covariates are shuffled. After seeing the current covariate and prediction, an adversary may still choose the label using the entire transcript. This *feature-random-order* model is stronger than standard random-order online optimization, which permutes complete loss functions and therefore labels as well (Garber et al., 2020; Sherman et al., 2021; Bernasconi et al., 2025). Nevertheless, for $r \ge 1$, the elementary unregularized Vovk–Azoury–Warmuth (VAW) forecaster achieves

$$\mathbb{E} \sup_u \mathrm{Reg}_T(u) \le B^2 \sum_{t=1}^{T} \frac{\min(r, t)}{t} = B^2 \{ r + r(H_T - H_r) \}, \qquad (1)$$

where r is the final design rank. The guarantee is scale-free, comparator-free, condition-number-free, and valid in an arbitrary Hilbert space after replacing dimension by algebraic Gram rank.

The main technical statement is stronger than an expectation bound. Write

$$V_t = \sum_{s \le t} x_s x_s^\top, \qquad h_t = x_t^\top V_t^\dagger x_t,$$

where $\dagger$ denotes the Moore–Penrose inverse. For every fixed multiset, the random-order leverage sum $S_T =$

Table 1: Doubly-uniform square-loss regret. In our model only covariates are randomly ordered and labels may be adaptive. Logarithms and confidence terms are suppressed.

covariate information	assumptions	regret
sequential, $d > 1$	adaptive	$\Theta(B^2 T)$
sequential	smooth density	$\widetilde{O}(B^2\sqrt{dC_{\mathrm{cov}}T})$
final Gram known	none	$B^2 d\log T + O(1)$
random order	none	$B^2 r\log(eT/r)$

$\sum_t h_t$ satisfies

$$\mathbb{E}e^{\lambda S_T} \le \prod_{t=1}^{T}\left[1+\frac{\min(r,t)}{t}(e^{\lambda}-1)\right] \quad (\lambda \ge 0). \quad (2)$$

The right side is the moment generating function of a Poisson-binomial random variable. A backward deletion argument proves the bound without martingale increments or matrix concentration. It also gives an exact expectation in terms of the average rank of a uniformly sampled subset.

Expected $O(d\log T)$ online leverage under random row order was observed by Cohen et al. (2020), based on a uniform-subset identity from Cohen et al. (2015). Our contribution is not that expectation estimate alone. It is the distributional domination, its sharpness, and the resulting exact-regret theorem under adaptive labels. Geometrically separated scalar covariates make S_T converge to the number of records in a random permutation, exactly the Bernoulli law in (2) for $r = 1$. Balanced repetitions of r coordinate vectors give $S_T = rH_{T/r}$ for every ordering. A separate lower bound shows that no algorithm can improve the order in (1), including its leading logarithmic constant asymptotically.

Our result also delineates nearby random-order work. Batch-to-online regression results randomize complete examples and give approximate regret (Dong and Yoshida, 2023; Sakaue and Yoshida, 2026). Random-order convex optimization assumes bounded domains and geometric regularity of the loss sequence (Garber et al., 2020; Sherman et al., 2021). Online row sampling controls adaptive streams using condition numbers or arithmetic restrictions (Goranci et al., 2026). None yields ordinary comparator-uniform regression regret when labels remain adaptive.

Contributions. First, we prove the Poisson-binomial Laplace domination and Bennett concentration of random-order online leverage. Second, we combine it with a pathwise VAW inequality to obtain high-probability doubly-uniform regret against adaptive labels. Third, we establish minimax sharpness and distributional sharpness. Fourth, we give exact rank-profile, Hilbert-space, and imperfect-shuffle extensions.

2 MODEL AND FORECASTER

Fix labeled covariates $\mathcal{X} = (x_1, \ldots, x_T)$ in $\mathbb{R}^d$. Repetitions and zero vectors are allowed. Nature draws a uniform permutation π of $[T]$. At round t, the learner observes $X_t = x_{\pi_t}$ and outputs $\widehat{y}_t$. An adversary then chooses $y_t \in [-B, B]$. The choice may depend on the learner, the revealed covariates, all earlier labels, and the current prediction. Our upper bound is pathwise in the labels, so it even allows dependence on the unrevealed permutation.

Let

$$V_t = \sum_{s=1}^{t} X_s X_s^\top, \qquad b_{t-1} = \sum_{s<t} y_s X_s.$$

The unregularized VAW prediction, clipped to the label interval, is

$$\widehat{y}_t = \mathrm{clip}_{[-B,B]}\big(X_t^\top V_t^\dagger b_{t-1}\big). \quad (3)$$

Equivalently, before clipping, it is the prediction of any minimizer of

$$\langle u, X_t\rangle^2 + \sum_{s<t}(\langle u, X_s\rangle - y_s)^2.$$

The prediction is unique even if the minimizer is not. A rank-revealing factorization inside the observed span uses $O(dr_t + r_t^2)$ storage and $O(dr_t + r_t^2)$ arithmetic on round t, where $r_t = \mathrm{rank}(V_t)$. These are standard VAW implementation costs and are not the source of the statistical improvement.

The following standard forward-algorithm inequality is the only regression-specific ingredient. We include a proof in Appendix A.

Lemma 1 (Pathwise VAW inequality)**.** *For every deterministic sequence (X_t, y_t) with $|y_t| \le B$, the predictor (3) satisfies, simultaneously for all $u \in \mathbb{R}^d$,*

$$\mathrm{Reg}_T(u) \le \sum_{t=1}^{T} y_t^2 h_t \le B^2 \sum_{t=1}^{T} h_t, \qquad h_t = X_t^\top V_t^\dagger X_t. \quad (4)$$

This inequality is implicit in the almost-unregularized analysis of Gaillard et al. (2019, Theorem 11). Its value here is that all randomness is confined to the covariate-only quantity $S_T = \sum_t h_t$.

3 RANDOM-ORDER ONLINE LEVERAGE

Let $r = \operatorname{rank}(V_T)$ and define, for $r \geq 1$,

$$\mu_{r,T} = \sum_{t=1}^{T} \frac{\min(r,t)}{t} = r + r(H_T - H_r), \tag{5}$$

$$v_{r,T} = \sum_{t=r+1}^{T} \frac{r}{t}\left(1 - \frac{r}{t}\right). \tag{6}$$

For $r = 0$, set both quantities to zero.

Theorem 2 (Poisson-binomial domination)**.** *For every fixed labeled covariate multiset and every $\lambda \geq 0$,*

$$\mathbb{E}_\pi e^{\lambda S_T} \leq e^{\lambda r} \prod_{t=r+1}^{T} \left(1 - \frac{r}{t} + \frac{r}{t} e^{\lambda}\right). \tag{7}$$

Thus S_T is dominated in Laplace transform by

$$Z_{r,T} = r + \sum_{t=r+1}^{T} Z_t, \qquad Z_t \sim \operatorname{Bernoulli}(r/t),$$
$$Z_t \perp\!\!\!\perp Z_s \quad (t \neq s).$$

Consequently, $\mathbb{E} S_T \leq \mu_{r,T}$ and, for every $s > 0$,

$$\mathbb{P}\{S_T \geq \mu_{r,T} + s\} \leq \exp\{-v_{r,T}\,\psi(s/v_{r,T})\}, \tag{8}$$

where $\psi(a) = (1+a)\log(1+a) - a$. If $v_{r,T} = 0$, the probability in (8) is zero. In particular, with probability at least $1 - \delta$,

$$S_T \leq \mu_{r,T} + \sqrt{2 v_{r,T} \log(1/\delta)} + \frac{1}{3}\log(1/\delta). \tag{9}$$

Proof idea. Generate the permutation backward. Given the labeled set A_t occupying the first t positions, choose its last index I_t uniformly and delete it. With $V(A) = \sum_{i \in A} x_i x_i^\top$,

$$\begin{aligned} \mathbb{E}[h_t \mid A_t] &= \frac{1}{t} \sum_{i \in A_t} x_i^\top V(A_t)^\dagger x_i \\ &= \frac{\operatorname{rank} V(A_t)}{t} \leq \frac{\min(r,t)}{t}. \end{aligned} \tag{10}$$

Each leverage lies in $[0,1]$. The exponential chord bound $e^{\lambda z} \leq 1 + z(e^\lambda - 1)$ and induction over the deletion recursion yield (7). Standard Bernoulli Bennett bounds give (8). Appendix B gives the full argument.

The backward identity also preserves the detailed geometry discarded by the rank envelope.

Proposition 3 (Exact rank profile)**.** *If A_t is a uniform labeled t-subset of $[T]$, then*

$$\mathbb{E}_\pi S_T = \sum_{t=1}^{T} \frac{1}{t} \mathbb{E}_{A_t} \operatorname{rank}\{x_i : i \in A_t\}. \tag{11}$$

The right side can be strictly below $\mu_{r,T}$.

Proposition 4 (Sharpness)**.** *Theorem 2 is sharp in two complementary senses.*

1. *For scalar covariates $x_i = M^i$, as $M \to \infty$, S_T converges almost surely to the number of upper records in π. Hence its moment generating function converges to (7) with $r = 1$.*
2. *If $T = rn$ and the multiset contains n copies of every coordinate vector $e_1, \ldots, e_r$, then $S_T = rH_n$ for every permutation.*

Part one shows that no uniformly smaller distributional proxy is possible, already in dimension one. Part two shows that the $r\log(T/r)$ scale does not arise from ill-conditioning or a rare ordering.

The contrast with adversarial arrival is already maximal for the first construction. If $x_i = M^i$ are presented in increasing order, every $h_t \to 1$ and $S_T \to T$ although the rank is one. Uniform order turns the same instance into a record process with mean H_T. Random arrival controls exactly the repeated scale expansions that defeat an unregularized sequential bound.

4 DOUBLY-UNIFORM REGRET

Combining two pathwise statements immediately handles adaptive labels.

Corollary 5 (Feature-random-order regret)**.** *For every fixed covariate multiset and every bounded, possibly adaptive label strategy, VAW satisfies*

$$\mathbb{E}_\pi \sup_{u \in \mathbb{R}^d} \operatorname{Reg}_T(u) \leq B^2 \mu_{r,T}, \tag{12}$$

$$\begin{aligned} \sup_u \operatorname{Reg}_T(u) \leq B^2 \big\{ &\mu_{r,T} \\ &+ \sqrt{2 v_{r,T} \log(1/\delta)} + \tfrac{1}{3}\log(1/\delta) \big\} \end{aligned} \tag{13}$$

with probability at least $1-\delta$ over the permutation. The probability statement is simultaneous over all possible bounded label realizations.

The supremum in (12) is harmless because Lemma 1 holds simultaneously for every comparator. Since $H_T - H_r \leq \log(T/r)$,

$$\mu_{r,T} \leq r\{1 + \log(T/r)\}.$$

No upper bound on $\|x_t\|$, $\|u\|$, the condition number, or the ambient dimension appears.

The result extends verbatim to a real Hilbert space $\mathcal{H}$. For a fixed horizon, all covariates lie in the finite-dimensional span $E = \operatorname{span}\{x_1, \ldots, x_T\}$. Projecting a comparator onto E preserves every prediction, and the proof runs inside E. Thus $r = \dim E$, equivalently the algebraic rank of the kernel Gram matrix. No kernel diagonal or effective-dimension bound is required.

Every finite exchangeable covariate sequence is an immediate corollary. Conditional on its unordered labeled multiset, its order is uniform. This includes i.i.d. covariates. Labels may still be selected after each covariate is seen, so this is not a stochastic-label assumption.

The guarantee can also be made simultaneous over all horizons. Let $r_k = \mathrm{rank}(V_k)$ and let $\mu_{r_k,k}$ and $v_{r_k,k}$ be defined as in (5)–(6). Set $L_k = \log\{\pi^2 k^2/(6\delta)\}$. A union bound applied conditionally on each unordered prefix gives, with probability at least $1-\delta$, for every $k \le T$,

$$\sup_u \mathrm{Reg}_k(u) \le B^2 \left\{ \mu_{r_k,k} + \sqrt{2v_{r_k,k}L_k} + \frac{L_k}{3} \right\}. \quad (14)$$

Thus neither the horizon nor the eventual rank is needed by the forecaster or by the data-dependent certificate.

4.1 Minimax lower bound

The logarithmic rate is unavoidable for every algorithm, even if the complete covariate schedule is revealed in advance.

Theorem 6 (Lower bound)**.** *Fix $1 \le r \le \min(d,T)$ and let $n = \lfloor T/r \rfloor$. For every online regression algorithm and every $\alpha > 0$, there is a rank-r covariate multiset and a bounded label strategy such that*

$$\mathbb{E} \sup_u \mathrm{Reg}_T(u) \ge \frac{2\alpha}{2\alpha+1} B^2 r \left(1 + \sum_{m=0}^{n-1} \frac{1}{2\alpha + m} \right). \quad (15)$$

The expectation includes the learner's randomness and the uniform covariate order. In particular, the minimax rate is $\Theta(B^2 r \log(eT/r))$. If $n \to \infty$ and then $\alpha \to \infty$ slowly enough, the lower bound is $(1-o(1))B^2 r \log n$.

The construction uses n copies of each coordinate vector and pads with zeros. Independently by coordinate, draw $p_j \sim \mathrm{Beta}(\alpha,\alpha)$ and labels with values $\pm B$ and mean $B(2p_j - 1)$. Posterior variance forces prediction cost, while the hindsight comparator is the sample mean. The exact conjugate calculation is in Appendix C. Averaging produces a deterministic adaptive label strategy, so the Bayesian device does not enlarge the adversary class.

5 IMPERFECT SHUFFLING

The distributional theorem quantifies how much randomness in the order is needed. Let U be uniform on the $T!$ permutations of the labeled occurrences, and let Q be any ordering law. Write $D = D_{\mathrm{KL}}(Q\|U) = \log(T!) - H(Q)$.

Corollary 7 (Entropy-deficient shuffles)**.** *For every $Q \ll U$,*

$$\mathbb{E}_Q S_T \le \mu_{r,T} + \sqrt{2v_{r,T}D} + \frac{D}{3}. \quad (16)$$

If $D_\infty(Q\|U) \le K$, then with Q-probability at least $1-\delta$,

$$S_T \le \mu_{r,T} + \sqrt{2v_{r,T}\{K + \log(1/\delta)\}} + \frac{K + \log(1/\delta)}{3}. \quad (17)$$

Both bounds transfer to regret after multiplication by B^2.

The proof applies the entropy variational formula to the sub-gamma consequence of (7). This provides a continuous interpolation between a uniform shuffle and a concentrated ordering law. Total entropy alone controls the mean. A worst-case density ratio is required for a high-probability transfer.

6 RELATED WORK AND LIMITS

The VAW and nonlinear ridge families originate in early work on online regression (Vovk, 2001; Azoury and Warmuth, 2001). Bartlett et al. (2015) characterized fixed-design minimax regret, and Gaillard et al. (2019) obtained the optimal $B^2 d \log T$ leading term when the design is known. For sequential designs, their scale-invariant bound contains a feature-dependent remainder. Chen et al. (2026) proved that this obstruction is fundamental in $d > 1$ for fully adaptive covariates, and derived square-root regret under smooth conditional covariate laws. Our logarithmic bound identifies random arrival of a fixed feature multiset as enough to remove the obstruction.

Random-order online optimization usually shuffles full loss functions. Logarithmic guarantees there use bounded domains, Lipschitzness, smoothness, and cumulative strong convexity (Garber et al., 2020; Sherman et al., 2021). General reductions compare random-order and i.i.d. online learning (Bernasconi et al., 2025). The batch-to-online method of Dong and Yoshida (2023) gives ϵ-approximate regression regret by randomizing complete examples. Recent refinements explain why its unsquared embedding error does not directly yield additive small-loss regret (Sakaue and Yoshida, 2026). In contrast, our labels can react to the current prediction and our regret is ordinary square-loss regret.

Online leverage is central to spectral row sampling. Cohen et al. (2020) note the expected $O(d \log T)$

random-order score sum. Under adaptive streams, Goranci et al. (2026) control related scores using an online condition number, or polynomially bounded integer entries for a horizon-only bound. Our contribution neither replaces those sparsification results nor their matrix concentration. It gives a sharp scalar distribution theorem under random order and turns it into an adaptive-label regret result.

Stochastic online regression permits arbitrary predictable covariates under realizability, sub-Gaussian noise, and bounded design and parameter assumptions (Ouhamma et al., 2021). Self-directed fixed-design regression asks the learner to choose an order under a random realizable target (Kontonis et al., 2023). These assumptions and objectives differ from doubly-uniform adversarial-label regret.

Three limitations are deliberate. Randomness is imposed on covariate arrival, so the theorem does not contradict the adaptive-design lower bound. Rank may equal T in a rich kernel space, in which case the bound becomes linear. Controlling effective dimension without a scale or regularization parameter would conflict with the invariance being sought. Finally, the condition-number-free guarantee is statistical and assumes exact arithmetic. Stable numerical rank decisions may still require a tolerance or regularization.

7 CONCLUSION

A random covariate order restores the same logarithmic doubly-uniform rate previously associated with advance design knowledge. The mechanism is exact: backward deletion converts each current leverage score into a rank-controlled Bernoulli surrogate. Because VAW regret is pathwise below the leverage sum, labels need not be randomized or stochastic. The resulting phase transition separates the role of feature arrival from assumptions on outcomes and supplies sharp expectation, concentration, and minimax guarantees without geometric normalization.

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A THE PATHWISE VAW INEQUALITY

We prove Lemma 1 by taking the zero-regularization limit of the forward algorithm. This also verifies that singular and rank-increasing designs cause no difficulty.

Fix $\lambda > 0$ and write

$$A_t = \lambda I + \sum_{s \le t} X_s X_s^\top, \qquad b_t = \sum_{s \le t} y_s X_s.$$

Consider the un-clipped prediction

$$p_{t,\lambda} = X_t^\top A_t^{-1} b_{t-1}, \qquad h_{t,\lambda} = X_t^\top A_t^{-1} X_t.$$

The quadratic potential

$$Q_t(u) = \lambda \|u\|^2 + \sum_{s \le t} (y_s - \langle u, X_s \rangle)^2$$

has minimum

$$\min_u Q_t(u) = \sum_{s \le t} y_s^2 - b_t^\top A_t^{-1} b_t. \tag{18}$$

Since $A_t = A_{t-1} + X_t X_t^\top$, the Sherman–Morrison formula, expressed in terms of A_t^{-1}, gives

$$b_{t-1}^\top A_{t-1}^{-1} b_{t-1} = b_{t-1}^\top A_t^{-1} b_{t-1} + \frac{p_{t,\lambda}^2}{1 - h_{t,\lambda}}.$$

Here $0 \le h_{t,\lambda} < 1$. Substituting this identity into (18) yields

$$\min Q_t - \min Q_{t-1} = \frac{\{(1 - h_{t,\lambda}) y_t - p_{t,\lambda}\}^2}{1 - h_{t,\lambda}}.$$

Direct expansion now shows

$$\begin{aligned}(y_t - p_{t,\lambda})^2 - (\min Q_t - \min Q_{t-1}) &= h_{t,\lambda} y_t^2 \\ &\quad - \frac{h_{t,\lambda} p_{t,\lambda}^2}{1 - h_{t,\lambda}} \\ &\le h_{t,\lambda} y_t^2. \end{aligned} \tag{19}$$

Summing and using $\min Q_0 = 0$ gives, for every u,

$$\sum_{t=1}^T (y_t - p_{t,\lambda})^2 \le \sum_{t=1}^T (y_t - \langle u, X_t \rangle)^2 + \lambda \|u\|^2 + \sum_{t=1}^T y_t^2 h_{t,\lambda}. \tag{20}$$

Let $V_t = \sum_{s \le t} X_s X_s^\top$. Both X_t and b_{t-1} lie in $\mathrm{range}(V_t)$. Spectral decomposition on this range gives

$$p_{t,\lambda} \longrightarrow X_t^\top V_t^\dagger b_{t-1}, \qquad h_{t,\lambda} \longrightarrow X_t^\top V_t^\dagger X_t.$$

There are finitely many rounds, so we may take $\lambda \downarrow 0$ in (20). Clipping a prediction to $[-B, B]$ cannot increase its square loss against $y_t \in [-B, B]$. This proves (4).

B LAPLACE DOMINATION AND CONCENTRATION

For a labeled index set $A \subseteq [T]$, define

$$V(A) = \sum_{i \in A} x_i x_i^\top, \qquad \ell_i(A) = x_i^\top V(A)^\dagger x_i.$$

If X_A is the matrix with rows $x_i^\top$, then $X_A V(A)^\dagger X_A^\top$ is the orthogonal projector onto the column space of X_A. Therefore

$$0 \le \ell_i(A) \le 1, \qquad \sum_{i \in A} \ell_i(A) = \operatorname{rank} V(A). \tag{21}$$

Generate a uniform permutation backward. Start from $A_T = [T]$. Conditional on A_t, select I_t uniformly in A_t, put $\pi_t = I_t$, and set $A_{t-1} = A_t \setminus \{I_t\}$. This construction is uniform because every deletion sequence has probability $1/T!$. Moreover,

$$h_t = \ell_{I_t}(A_t).$$

Equation (21) proves the conditional identity

$$\mathbb{E}[h_t \mid A_t] = \frac{\operatorname{rank} V(A_t)}{t}.$$

We next retain dependence across all deletion steps. For $A \subseteq [T]$, let

$$F(A) = \mathbb{E}\left[\exp\left\{ \lambda \sum_{s=1}^{|A|} h_s \right\} \,\middle|\, A_{|A|} = A \right].$$

Backward deletion gives the exact recursion

$$F(A) = \frac{1}{|A|} \sum_{i \in A} e^{\lambda \ell_i(A)} F(A \setminus \{i\}). \tag{22}$$

Fix the final rank r. We prove by induction on $t = |A|$ that, for every reachable A,

$$F(A) \le \prod_{s=1}^t \left[1 + \frac{\min(r, s)}{s} (e^\lambda - 1) \right]. \tag{23}$$

The empty-set case is immediate. The inductive bound on every child in (22) is the same deterministic product through $t-1$. For $z \in [0, 1]$ and $\lambda \ge 0$, convexity gives

$$e^{\lambda z} \le 1 + z(e^\lambda - 1).$$

Thus (21) and $\operatorname{rank} V(A) \le \min(r, t)$ show

$$\frac{1}{t} \sum_{i \in A} e^{\lambda \ell_i(A)} \le 1 + \frac{\min(r, t)}{t} (e^\lambda - 1).$$

This closes the induction. At $A = [T]$, the product in (23) is exactly the right side of (7).

Differentiating the inequality at $\lambda = 0$ from the right gives $\mathbb{E}S_T \le \mu_{r,T}$. For concentration, let $Z_{r,T} = r + \sum_{t=r+1}^{T} Z_t$ be the independent Bernoulli proxy in Theorem 2. The standard Bennett bound for centered Bernoulli variables gives

$$\log \mathbb{E}e^{\lambda(Z_{r,T}-\mu_{r,T})} \le v_{r,T}(e^{\lambda} - 1 - \lambda), \qquad \lambda \ge 0. \quad (24)$$

Laplace domination implies the same upper bound for $S_T - \mu_{r,T}$. Optimizing Chernoff's inequality at $\lambda = \log(1 + s/v_{r,T})$ proves (8). Finally,

$$e^{\lambda} - 1 - \lambda \le \frac{\lambda^2}{2(1-\lambda/3)}, \qquad 0 \le \lambda < 3,$$

and the usual sub-gamma inversion proves (9).

B.1 Exact expectation and sharpness

For every t, the backward set A_t is a uniform t-subset. Taking expectations in (10) and summing proves Proposition 3.

For the second part of Proposition 4, suppose coordinate j has appeared k times including the current round. The current Gram matrix has eigenvalue k in direction e_j, so the current leverage is $1/k$. Every coordinate therefore contributes $\sum_{k=1}^{n} 1/k = H_n$, regardless of interleaving.

For the first part, take scalar $x_i = M^i$ with distinct indices i. If π_t is the largest index among $\pi_1, \dots, \pi_t$, then

$$h_t = \frac{M^{2\pi_t}}{\sum_{s \le t} M^{2\pi_s}} \longrightarrow 1.$$

Otherwise the denominator contains a term larger by at least a factor M^2, and $h_t \to 0$. Hence S_T converges pointwise on the finite permutation space to the number of upper records. The relative insertion ranks of a uniform permutation are independent, with the rank at time t uniform on $[t]$. Its record indicators are therefore independent Bernoulli variables with means $1/t$. Bounded convergence proves convergence of every moment generating function to (7) for $r = 1$.

B.2 Hilbert-space reduction

Let $x_1, \dots, x_T$ belong to a real Hilbert space $\mathcal{H}$ and let $E = \operatorname{span}\{x_1, \dots, x_T\}$. This subspace is finite dimensional and closed. If P_E is its orthogonal projector, then $\langle u, x_t \rangle = \langle P_E u, x_t \rangle$ for every $u \in \mathcal{H}$. The prediction problem and comparator loss therefore depend only on $P_E u$. Choose an orthonormal basis of E and identify it with $\mathbb{R}^r$. The VAW update, all leverage scores, and every proof above are unchanged. Since the nonzero eigenvalues of the operator Gram matrix and the $T \times T$ kernel Gram matrix agree, r is the algebraic kernel Gram rank.

B.3 Simultaneous horizons

Fix $k \le T$ and condition on the labeled set A_k occupying the first k positions. Its internal order is uniform, and its rank is the now deterministic value r_k. Theorem 2 and Lemma 1, applied with failure probability $\delta_k = 6\delta/(\pi^2 k^2)$, show that (14) fails at this k with conditional probability at most δ_k. The same bound holds after removing the conditioning. Since $\sum_{k \ge 1} \delta_k = \delta$, a union bound proves (14) simultaneously for all $k \le T$.

C THE MINIMAX LOWER BOUND

We prove Theorem 6. Let $n = \lfloor T/r \rfloor$. Use n copies of each of $e_1, \dots, e_r$ and $T - rn$ zero covariates. Give the zero covariates label zero. We may reveal the complete future covariate schedule to the learner, since this only makes the problem easier.

For every coordinate j, draw independently

$$p_j \sim \mathrm{Beta}(\alpha, \alpha).$$

At each occurrence of e_j, independently conditional on p_j, set

$$Y = B(2W - 1), \qquad W \sim \mathrm{Bernoulli}(p_j).$$

Information from other coordinates is independent of p_j. Consequently, after m earlier labels on coordinate j, the posterior mean of Y minimizes conditional expected square loss for the next label. For every learner, its expected loss on that occurrence is at least

$$\begin{aligned} \mathbb{E}\operatorname{Var}(Y \mid Y_{1:m}) &= \mathbb{E}[4B^2 p_j(1-p_j)] \\ &+ \mathbb{E}\operatorname{Var}(B(2p_j - 1) \mid Y_{1:m}). \end{aligned} \quad (25)$$

The two terms have closed forms. A symmetric beta variable satisfies

$$\mathbb{E}[4B^2 p_j(1-p_j)] = \frac{2\alpha}{2\alpha+1} B^2. \quad (26)$$

For completeness, $\operatorname{Var}(p_j) = 1/[4(2\alpha+1)]$. If K_m is the number of positive labels among m, the posterior mean of p_j is $(\alpha + K_m)/(2\alpha + m)$, while the beta-binomial variance is

$$\operatorname{Var}(K_m) = \frac{m(2\alpha+m)}{4(2\alpha+1)}.$$

The law of total variance therefore gives

$$\mathbb{E}\operatorname{Var}(B(2p_j - 1) \mid Y_{1:m}) = \frac{2\alpha}{(2\alpha+1)(2\alpha+m)} B^2. \quad (27)$$

Summing (25) over $m = 0, \dots, n-1$ lower bounds the learner's loss on coordinate j. The best constant prediction in hindsight is the sample mean. Conditional on p_j, its residual sum of squares has expectation $(n-1)4B^2 p_j(1-p_j)$. After averaging, subtracting this comparator loss leaves one copy of (26) plus the sum of (27). Summing over the r independent coordinates proves (15).

This argument directly covers randomized learners by including their private randomness in the outer expectation. It uses a randomized label strategy only as an averaging device. Pre-sample p_j and an infinite label sequence for every coordinate. Each seed specifies a deterministic, nonanticipating strategy whose next label depends only on the current coordinate and its occurrence count. Since the average over seeds satisfies (15), at least one deterministic seed does as well.

Taking $\alpha = 1$ reduces (15) to

$$\frac{2}{3}B^2 r H_{n+1},$$

which is a universal-constant lower bound of order $B^2 r \log(eT/r)$. If $n \to \infty$, choose for example $\alpha = \max\{1, \lceil \log n \rceil\}$. Then

$$\frac{2\alpha}{2\alpha+1} \to 1, \qquad \sum_{m=0}^{n-1} \frac{1}{2\alpha+m} = (1-o(1))\log n.$$

This proves the leading-constant statement.

D ENTROPY TRANSPORT

We prove Corollary 7. Equations (7) and (24) imply, under the uniform law U,

$$\log \mathbb{E}_U e^{\lambda(S_T - \mu_{r,T})} \le \frac{v_{r,T}\lambda^2}{2(1-\lambda/3)}, \qquad 0 < \lambda < 3. \quad (28)$$

The entropy variational inequality gives

$$\lambda(\mathbb{E}_Q S_T - \mu_{r,T}) \le D_{\mathrm{KL}}(Q\|U) + \log \mathbb{E}_U e^{\lambda(S_T - \mu_{r,T})}.$$

If $v_{r,T} > 0$ and $D = D_{\mathrm{KL}}(Q\|U)$, choose

$$\lambda = \frac{\sqrt{2D/v_{r,T}}}{1 + \frac{1}{3}\sqrt{2D/v_{r,T}}}.$$

Substitution proves (16). The cases $D = 0$ or $v_{r,T} = 0$ follow by continuity. Since U assigns mass $1/T!$ to every labeled permutation, $D = \log(T!) - H(Q)$.

For the tail, $D_\infty(Q\|U) \le K$ means $Q(E) \le e^K U(E)$ for every event E. Apply (9) under U with failure probability δe^{-K}. This gives (17). Lemma 1 transfers both leverage statements to regret.

E ADDITIONAL COMPARISONS

We record protocol details that distinguish Corollary 5 from nearby results.

Complete-loss random order. In the standard random-order model, an adversary fixes loss functions $f_1, \ldots, f_T$ and a uniform permutation decides which full loss appears at each round. For square loss, fixing $f_t(u) = (\langle u, x_t \rangle - y_t)^2$ couples the label to the shuffled covariate. Our model shuffles x_t alone and permits y_t to be selected after $\widehat{y}_t$. It therefore cannot be reduced to a complete-loss random-order theorem.

Approximate versus ordinary regret. The ϵ-approximate regret of Dong and Yoshida (2023) compares online loss with $(1+\epsilon)$ times the offline loss. For regression, this introduces a multiplicative slack depending on the comparator's loss. Corollary 5 is additive ordinary regret and is uniform even when the comparator loss is arbitrarily large.

Adaptive versus random-order leverage. For an arbitrary row order, the unregularized online leverage sum can depend on scale ratios. Recent adaptive-stream sparsification bounds retain this dependence through an online condition number unless entries obey an arithmetic restriction (Goranci et al., 2026). Proposition 4 shows why no such quantity is present under uniform order: even arbitrarily separated scales produce a universal record law.